CSE-361:Compiler Design

Parsing : Part-II

Predictive Parsing

- The goal is to construct a top-down parser that never backtracks
- Always leftmost derivations
- We must transform a grammar in two ways:
  - eliminate left recursion
  - perform left factoring
- These rules eliminate most common causes for backtracking although they do not guarantee a completely backtrack-free parsing

Left Recursion: Infinite Looping problem

- **A grammar is left-recursive if it has a non-terminal A, such that there is a derivation :**

  $$A \overset{+}{\Rightarrow} A\alpha, \quad \text{for some string } \alpha.$$

- **Top-Down parsing can't reconcile this type of grammar, since it could consistently make choice which wouldn't allow termination.**

  $A \Rightarrow A\alpha \Rightarrow A\alpha\alpha \Rightarrow A\alpha\alpha\alpha \ldots$ etc. $\quad A \rightarrow A\alpha \mid \beta$

- **So we have to convert our left-recursive grammar into an equivalent grammar which is not left-recursive.**

- The left-recursion may appear in a single step of the derivation (*immediate left-recursion*), or may appear in more than one step of the derivation.

Immediate Left Recursion

- $A \rightarrow A\ \alpha \mid \beta$ where β does not start with A

$\Downarrow$ eliminate immediate left recursion

$A \rightarrow \beta\ A'$ where A' is a new nonterminal

$A' \rightarrow \alpha\ A' \mid \varepsilon$ an equivalent grammar

More General (but still immediate):

$$A \rightarrow A\alpha_1 \mid A\alpha_2 \mid A\alpha_3 \mid \ldots \mid \beta_1 \mid \beta_2 \mid \beta_3 \mid \ldots$$

Transform into:

$$A \rightarrow \beta_1 A' \mid \beta_2 A' \mid \beta_3 A' \mid \ldots$$

$$A' \rightarrow \alpha_1 A' \mid \alpha_2 A' \mid \alpha_3 A' \mid \ldots \mid \varepsilon$$

Immediate Left Recursion Elimination: Example

Our Example :

$E \rightarrow E + T \mid T$ $\longrightarrow$ $\begin{cases} E \rightarrow TE' \\ E' \rightarrow + TE' \mid \in \end{cases}$

$T \rightarrow T * F \mid F$ $\longrightarrow$ $\begin{cases} T \rightarrow FT' \\ T' \rightarrow * FT' \mid \in \end{cases}$

$F \rightarrow (E) \mid id$ $\longrightarrow$ $F \rightarrow (E) \mid id$

Left Recursion in More Than One Step

- A grammar cannot be immediately left-recursive, but it still can be left-recursive.
- By just eliminating the immediate left-recursion, we may not get a grammar which is not left-recursive.

Example:

$S \rightarrow A\underline{f} \mid \underline{b}$

$A \rightarrow A\underline{c} \mid S\underline{d} \mid \underline{e}$

Is A left recursive? Yes.

Is S left recursive? Yes, but not immediate left recursion. $S \Rightarrow A\underline{f} \Rightarrow S\underline{df}$

Left Recursion in More Than One Step: Elimination

Approach:

Look at the rules for S only (ignoring other rules)... No left recursion.

Look at the rules for A...

Do any of A's rules start with S? Yes.

$A \rightarrow S\underline{d}$

Get rid of the S. Substitute in the righthand sides of S.

$A \rightarrow A\underline{fd} \mid \underline{bd}$

The modified grammar:

$S \rightarrow A\underline{f} \mid \underline{b}$

$A \rightarrow A\underline{e} \mid A\underline{fd} \mid \underline{bd} \mid \underline{e}$

Now eliminate immediate left recursion involving A.

$S \rightarrow A\underline{f} \mid \underline{b}$

$A \rightarrow \underline{bd}A' \mid \underline{e}A'$

$A' \rightarrow \underline{e}A' \mid \underline{fd}A' \mid \underline{\varepsilon}$

Left Recursion in More Than One Step: Elimination

The Original Grammar:

$S \rightarrow A\underline{f} \mid \underline{b}$

$A \rightarrow A\underline{c} \mid S\underline{d} \mid B\underline{e}$

$B \rightarrow A\underline{g} \mid S\underline{h} \mid \underline{k}$

So Far:

$S \rightarrow A\underline{f} \mid \underline{b}$

$A \rightarrow \underline{bd}A' \mid B\underline{e}A'$

$A' \rightarrow \underline{c}A' \mid \underline{fd}A' \mid \varepsilon$

Left Recursion in More Than One Step: Elimination

The Original Grammar:

$S \rightarrow Af \mid b$

$A \rightarrow Ac \mid Sd \mid Be$

$B \rightarrow Ag \mid Sh \mid k$

So Far:

$S \rightarrow Af \mid b$

$A \rightarrow bdA' \mid BeA'$

$A' \rightarrow cA' \mid fdA' \mid \varepsilon$

$B \rightarrow Ag \mid Sh \mid k$

Look at the B rules next; Does any righthand side start with "S"?

Left Recursion in More Than One Step: Elimination

The Original Grammar:

$S \rightarrow Af \mid b$

$A \rightarrow Ac \mid Sd \mid Be$

$B \rightarrow Ag \mid Sh \mid k$

So Far:

$S \rightarrow Af \mid b$

$A \rightarrow bdA' \mid BeA'$

$A' \rightarrow cA' \mid fdA' \mid \varepsilon$

$B \rightarrow Ag \mid Afh \mid bh \mid k$

Substitute, using the rules for "S"

$Af\ldots \mid b\ldots$

Left Recursion in More Than One Step: Elimination

The Original Grammar:

$S \rightarrow Af \mid b$

$A \rightarrow Ac \mid Sd \mid Be$

$B \rightarrow Ag \mid Sh \mid k$

So Far:

$S \rightarrow Af \mid b$

$A \rightarrow bdA' \mid BeA'$

$A' \rightarrow cA' \mid fdA' \mid \varepsilon$

$B \rightarrow Ag \mid Afh \mid bh \mid k$

Does any righthand side start with "A"?

Left Recursion in More Than One Step: Elimination

The Original Grammar:

$S \rightarrow Af \mid b$

$A \rightarrow Ac \mid Sd \mid Be$

$B \rightarrow Ag \mid Sh \mid k$

So Far:

$S \rightarrow Af \mid b$

$A \rightarrow bdA' \mid BeA'$

$A' \rightarrow cA' \mid fdA' \mid \varepsilon$

$B \rightarrow bdA'g \mid BeA'g \mid Afh \mid bh \mid k$

Substitute, using the rules for "A"

$bdA'... \mid BeA'...$

Left Recursion in More Than One Step: Elimination

The Original Grammar:

$S \rightarrow Af \mid b$

$A \rightarrow Ac \mid Sd \mid Be$

$B \rightarrow Ag \mid Sh \mid k$

So Far:

$S \rightarrow Af \mid b$

$A \rightarrow bdA' \mid BeA'$

$A' \rightarrow cA' \mid fdA' \mid \varepsilon$

$B \rightarrow bdA'g \mid BeA'g \mid bdA'fh \mid BeA'fh \mid bh \mid k$

Substitute, using the rules for "A"

$bdA'\ldots \mid BeA'\ldots$

Left Recursion in More Than One Step: Elimination

The Original Grammar:

$S \rightarrow Af \mid b$

$A \rightarrow Ac \mid Sd \mid Be$

$B \rightarrow Ag \mid Sh \mid k$

So Far:

$S \rightarrow Af \mid b$

$A \rightarrow bdA' \mid BeA'$

$A' \rightarrow cA' \mid fdA' \mid \varepsilon$

$B \rightarrow bdA'g \mid BeA'g \mid bdA'fh \mid BeA'fh \mid bh \mid k$

Finally, eliminate any immediate Left recursion involving "B"

Next Form

$S \rightarrow Af \mid b$

$A \rightarrow bdA' \mid BeA'$

$A' \rightarrow cA' \mid fdA' \mid \varepsilon$

$B \rightarrow bdA'gB' \mid bdA'fhB' \mid bhB' \mid kB'$

$B' \rightarrow eA'gB' \mid eA'fhB' \mid \varepsilon$

Left Recursion in More Than One Step: Elimination

The Original Grammar:

$S \rightarrow Af \mid b$

$A \rightarrow Ac \mid Sd \mid Be \mid C$

$B \rightarrow Ag \mid Sh \mid k$

$C \rightarrow Bkm A \mid AS \mid j$

If there is another nonterminal, then do it next.

So Far:

$S \rightarrow Af \mid b$

$A \rightarrow bdA' \mid BeA' \mid CA'$

$A' \rightarrow cA' \mid fdA' \mid \varepsilon$

$B \rightarrow bdA'gB' \mid bdA'fhB' \mid bhB' \mid kB' \mid CA'gB' \mid CA'fhB'$

$B' \rightarrow eA'gB' \mid eA'fhB' \mid \varepsilon$

Algorithm for Eliminating Left Recursion

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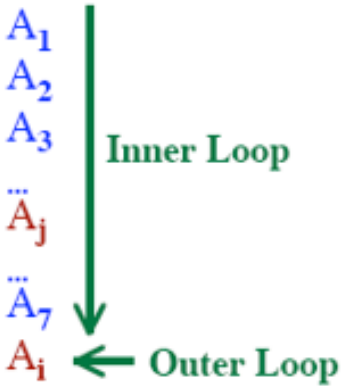
Assume the nonterminals are ordered A1, A2, A3,...
        (In the example: S, A, B)
for each nonterminal Ai (for i = 1 to N) do
  for each nonterminal Aj (for j = 1 to i-1) do
    Let Aj → β1 | β2 | β3 | ... | βN be all the rules for Aj
    if there is a rule of the form
       Ai → Ajα
    then replace it by
       Ai → β1α | β2α | β3α | ...  | βNα
    endIf
  endFor
  Eliminate immediate left recursion
        among the Ai rules
endFor
```



Left Factoring: Common Prefix Problem

Problem : Uncertain which of 2 rules to choose:

stmt → if *expr* then *stmt* else *stmt*

| if *expr* then *stmt*

When do you know which one is valid ?

What's the general form of *stmt* ?

$A \rightarrow \alpha\beta_1 \mid \alpha\beta_2$

α : if *expr* then *stmt*

β_1: else *stmt* β_2 : $\in$

Transform to:

$A \rightarrow \alpha A'$

$A' \rightarrow \beta_1 \mid \beta_2$

EXAMPLE:

stmt → if *expr* then *stmt* *rest*

rest → else *stmt* | $\in$

Left Factoring : Example

$A \rightarrow abB \mid aB \mid cdg \mid cdeB \mid cdfB$

$\Downarrow$

$A \rightarrow aA' \mid cdg \mid cdeB \mid cdfB$

$A' \rightarrow bB \mid B$

$\Downarrow$

$A \rightarrow aA' \mid cdA''$

$A' \rightarrow bB \mid B$

$A'' \rightarrow g \mid eB \mid fB$

Left Factoring : Example

$A \rightarrow ad \mid a \mid ab \mid abc \mid b$

$\Downarrow$

$A \rightarrow aA' \mid b$

$A' \rightarrow d \mid \varepsilon \mid b \mid bc$

$\Downarrow$

$A \rightarrow aA' \mid b$

$A' \rightarrow d \mid \varepsilon \mid bA''$

$A'' \rightarrow \varepsilon \mid c$

Reading Materials

- Chapter -4 of your Text book:
  - Compilers: Principles, Techniques, and Tools

The End